

17.5 A One-Loop Calculation in Background Field Gauge

As an exercise, we are now going to calculate the one-loop renormalization factor for the gauge coupling constant in a general non-Abelian gauge theory. As we will see in the next chapter, this provides an essential input in so-called ‘renormalization group’ calculations of physical processes at high energy; the results we obtain here will be used there to demonstrate the asymptotic freedom of non-Abelian gauge theories.

The method to be employed here is somewhat novel. Usually, one considers the effective action in a spacetime-dependent background gauge field, and calculates the terms quadratic in this field, extracting a factor $(q_\mu A_{\alpha\nu} - q_\nu A_{\alpha\mu})^2$ (where q is the gauge field four-momentum), and only then isolating the logarithmic divergence by setting $q = 0$ in the coefficient of this factor. Instead we shall follow the much simpler course of taking the gauge field to be spacetime-independent from the beginning. In this case the terms in the effective action that are quadratic or cubic in the gauge field of course vanish, but there is a non-vanishing quartic term that is ultraviolet divergent, and which can be used to calculate the coupling constant renormalization factor $(1 + L_A)^{-1/2}$. In this way our one-loop calculation becomes a matter of simple matrix algebra. Note that this procedure can only work in background field gauge; otherwise there would be independent logarithmic divergences in the parts of the effective action that are quartic and quadratic in the background gauge field.

With this motivation, we turn to the calculation of the one-loop effective action in a background field for which $A_{\alpha\mu}$ is constant and $\psi = \omega = \omega^* = 0$. For such a background field, the full modified Lagrangian is¹

$$\mathcal{L}_{\text{MOD}} = \mathcal{L} + \mathcal{L}_f + \mathcal{L}_{GH}, \quad (17.5.1)$$

$$\begin{aligned} \mathcal{L} = & -\frac{1}{4} (F_{\alpha\mu\nu} + \bar{D}_\mu A'_{\alpha\nu} - \bar{D}_\nu A'_{\alpha\mu} + C_{\alpha\beta\gamma} A'_{\beta\mu} A'_{\gamma\nu})^2 \\ & - \bar{\psi}' (\not{D} - it_\alpha A'_\alpha + m) \psi', \end{aligned} \quad (17.5.2)$$

$$\mathcal{L}_f = -\frac{1}{2\xi} f_\alpha f_\alpha = -\frac{1}{2\xi} (\bar{D}_\mu A'^\mu_\alpha)^2, \quad (17.5.3)$$

$$\mathcal{L}_{GH} = -(\bar{D}_\mu \omega'^*_\alpha) (\bar{D}^\mu \omega'_\alpha - C_{\alpha\beta\gamma} \omega'_\beta A'^\mu_\gamma). \quad (17.5.4)$$

One-loop graphs for vacuum–vacuum amplitudes are calculated from the part of the action that is quadratic in the quantum fields $A', \psi', \omega', \omega'^*$ over

¹See Eqs. (??), (??), and (??). We are here specializing to the case of matter fields forming a multiplet of spin 1/2 fermions. The squares in $\mathcal{L}$ and $\mathcal{L}_f$ include obvious index contractions.

which one integrates. Keeping only such quadratic terms, we have

$$\begin{aligned}\mathcal{L}_{\text{QUAD}} = & -\frac{1}{4}(\bar{D}_\mu A'_{\alpha\nu} - \bar{D}_\nu A'_{\alpha\mu})^2 - \frac{1}{2}F_\alpha^{\mu\nu}C_{\alpha\beta\gamma}A'_{\beta\mu}A'_{\gamma\nu} \\ & - \bar{\psi}'(\not{D} + m)\psi' - \frac{1}{2\xi}(\bar{D}_\mu A'^\mu_\alpha)^2 - (\bar{D}_\mu \omega'^*_\alpha)(\bar{D}^\mu \omega'_\alpha).\end{aligned}\quad (17.5.5)$$

The corresponding action may be put in the general quadratic form:

$$\begin{aligned}I_{\text{QUAD}} \equiv & \int d^4x \mathcal{L}_{\text{QUAD}} \\ = & -\frac{1}{2} \int d^4x d^4y A'^\mu_\alpha(x) A'^\nu_\beta(y) \mathcal{D}^A_{x\alpha\mu, y\beta\nu} - \int d^4x d^4y \bar{\psi}'_k(x) \psi'_\ell(y) \mathcal{D}^\psi_{xk, y\ell} \\ & - \int d^4x d^4y \omega'^*_\alpha(x) \omega'_\beta(y) \mathcal{D}^\omega_{x\alpha, y\beta},\end{aligned}\quad (17.5.6)$$

with

$$\begin{aligned}\mathcal{D}^A_{x\alpha\mu, y\beta\nu} \equiv & \eta_{\mu\nu} \left(-\delta_{\gamma\alpha} \frac{\partial}{\partial x_\lambda} + C_{\gamma\delta\alpha} A^\lambda_\delta(x) \right) \left(-\delta_{\gamma\beta} \frac{\partial}{\partial y^\lambda} + C_{\gamma\epsilon\beta} A_{\epsilon\lambda}(y) \right) \delta^4(x-y) \\ & - \left(-\delta_{\gamma\alpha} \frac{\partial}{\partial x^\nu} + C_{\gamma\delta\alpha} A_{\delta\nu}(x) \right) \left(-\delta_{\gamma\beta} \frac{\partial}{\partial y^\mu} + C_{\gamma\epsilon\beta} A^\mu_\epsilon(y) \right) \delta^4(x-y) \\ & + F_{\gamma\mu\nu}(x) C_{\gamma\alpha\beta} \delta^4(x-y) \\ & + \frac{1}{\xi} \left(-\delta_{\gamma\alpha} \frac{\partial}{\partial x^\mu} + C_{\gamma\delta\alpha} A_{\delta\mu}(x) \right) \left(-\delta_{\gamma\beta} \frac{\partial}{\partial y^\nu} + C_{\gamma\epsilon\beta} A_{\epsilon\nu}(y) \right) \delta^4(x-y),\end{aligned}\quad (17.5.7)$$

$$\mathcal{D}^\psi_{xk, y\ell} \equiv \left(-\gamma^\mu \frac{\partial}{\partial y^\mu} - it_\alpha \not{A}_\alpha(y) + m \right)_{k\ell} \delta^4(x-y), \quad (17.5.8)$$

$$\mathcal{D}^\omega_{x\alpha, y\beta} \equiv \left(-\delta_{\gamma\alpha} \frac{\partial}{\partial x^\lambda} + C_{\gamma\delta\alpha} A_{\delta\lambda}(x) \right) \left(-\delta_{\gamma\beta} \frac{\partial}{\partial y^\lambda} + C_{\gamma\epsilon\beta} A^\lambda_\epsilon(y) \right) \delta^4(x-y). \quad (17.5.9)$$

(The minus signs in front of $\partial/\partial x$ and $\partial/\partial y$ drop out when we integrate by parts.)

The one-loop contribution to the effective action is given (as in Section 16.2) by

$$\begin{aligned}\exp(i\Gamma^{(1\text{ loop})}[A]) \propto & \int_{1\text{PI}} \left(\prod dA' \right) \left(\prod d\psi' \right) \left(\prod d\bar{\psi}' \right) \left(\prod d\omega' \right) \left(\prod d\omega'^* \right) \\ & \times \exp(iI_{\text{QUAD}}[A', \psi', \bar{\psi}', \omega', \omega'^*; A]) \\ \propto & (\text{Det } \mathcal{D}^A)^{-1/2} (\text{Det } \mathcal{D}^\psi)^{+1} (\text{Det } \mathcal{D}^\omega)^{+1}.\end{aligned}\quad (17.5.10)$$

(The exponents $-1/2$ and $+1$ appear because A' is a real boson field, while ψ' , $\bar{\psi}'$, ω' , and ω'^* are distinct fermionic fields.) The calculation of such determinants is generally not an easy task. However, it becomes much simpler

in the case of constant external fields, where the $\mathcal{D}$ s can be diagonalized by passing to momentum space.

Let us therefore now consider the case of constant background field $A_{\alpha\mu}$. For non-Abelian gauge theories such a constant field *cannot* be removed by a gauge transformation, as shown by the non-zero values of various gauge-covariant fields

$$F_{\alpha\mu\nu} = C_{\alpha\beta\gamma} A_{\beta\mu} A_{\gamma\nu}, \quad (17.5.11)$$

$$D_\lambda F_{\delta\mu\nu} = C_{\delta\epsilon\alpha} C_{\alpha\beta\gamma} A_{\epsilon\lambda} A_{\beta\mu} A_{\gamma\nu}, \quad (17.5.12)$$

and so on. Lorentz and gauge invariance tell us how to express the part of $\Gamma[A]$ of a given dimensionality as the integral of a finite number of local functions of $F_{\alpha\mu\nu}$, $D_\lambda F_{\alpha\mu\nu}$, etc.; the coefficients of the terms in this expression can be inferred by comparing the contribution that these terms make to $\Gamma[A]$ for constant background field $A_{\alpha\mu}$ with the results of a perturbative expansion.

We transform each of the ‘matrices’ $\mathcal{D}^A$, $\mathcal{D}^\psi$, and $\mathcal{D}^\omega$ to a momentum basis by the usual normalized Fourier transform

$$\mathcal{D}_{q\cdots,p\cdots} = \int \frac{d^4x}{(2\pi)^2} e^{-iq\cdot x} \int \frac{d^4y}{(2\pi)^2} e^{ip\cdot y} \mathcal{D}_{x\cdots,y\cdots}. \quad (17.5.13)$$

With A constant, this gives

$$\mathcal{D}_{q\cdots,p\cdots} = \delta^4(p - q) \mathcal{M}_{\cdots,\cdots}(q), \quad (17.5.14)$$

where $\cdots$ denotes discrete indices, and the $\mathcal{M}$ are finite q -dependent matrices

$$\begin{aligned} \mathcal{M}_{\alpha\mu,\beta\nu}^A(q) = & \eta_{\mu\nu} (-iq^\lambda \delta_{\gamma\alpha} + A_\delta^\lambda C_{\gamma\delta\alpha}) (iq_\lambda \delta_{\gamma\beta} + A_{\epsilon\lambda} C_{\gamma\epsilon\beta}) \\ & - (-iq_\nu \delta_{\gamma\alpha} + A_{\delta\nu} C_{\gamma\delta\alpha}) (iq_\mu \delta_{\gamma\beta} + A_\epsilon^\mu C_{\gamma\epsilon\beta}) + F_{\gamma\mu\nu} C_{\gamma\alpha\beta} \\ & + \frac{1}{\xi} (-iq_\mu \delta_{\gamma\alpha} + A_{\delta\mu} C_{\gamma\delta\alpha}) (iq_\nu \delta_{\gamma\beta} + A_{\epsilon\nu} C_{\gamma\epsilon\beta}) + \epsilon \text{ terms}, \end{aligned} \quad (17.5.15)$$

$$\mathcal{M}_{k\ell}^\psi(q) = (i\not{q} - it_\alpha \not{A}_\alpha + m)_{k\ell} + \epsilon \text{ terms}, \quad (17.5.16)$$

$$\mathcal{M}_{\alpha\beta}^\omega(q) = (-iq_\lambda \delta_{\gamma\alpha} + A_{\delta\lambda} C_{\gamma\delta\alpha}) (iq^\lambda \delta_{\gamma\beta} + A_\epsilon^\lambda C_{\gamma\epsilon\beta}) + \epsilon \text{ terms}, \quad (17.5.17)$$

with $F_{\gamma\mu\nu}$ given by Eq. (17.5.11). From Eq. (17.5.10) we have then

$$\begin{aligned} i\Gamma^{(1 \text{ loop})}[A] = & -\frac{1}{2} \ln \text{Det } \mathcal{D}^A + \ln \text{Det } \mathcal{D}^\psi + \ln \text{Det } \mathcal{D}^\omega \\ = & -\frac{1}{2} \text{Tr} \ln \mathcal{D}^A + \text{Tr} \ln \mathcal{D}^\psi + \text{Tr} \ln \mathcal{D}^\omega \\ = & \delta^4(p - p) \int d^4q \left[-\frac{1}{2} \text{tr} \ln \mathcal{M}^A(q) + \text{tr} \ln \mathcal{M}^\psi(q) + \text{tr} \ln \mathcal{M}^\omega(q) \right]. \end{aligned} \quad (17.5.18)$$

We denote traces by ‘tr’ instead of ‘Tr’ in the last line of Eq. (17.5.18) (and from now on in this section) to indicate that these are the usual traces of finite matrices rather than of integral operators.

Since we are aiming here at a calculation of the infinite factor L_A multiplying FF terms in the effective action, let’s isolate the term in (17.5.18) that is of fourth order in the background field A . For this purpose, it is convenient to divide each $\mathcal{M}$ into terms $\mathcal{M}_n$ containing $n = 0, 1$, or 2 factors of A :

$$\mathcal{M} = \mathcal{M}_0 + \mathcal{M}_1 + \mathcal{M}_2. \quad (17.5.19)$$

It is then elementary algebra to show that the term in (17.5.18) of fourth order in $A_{\alpha\mu}$ is

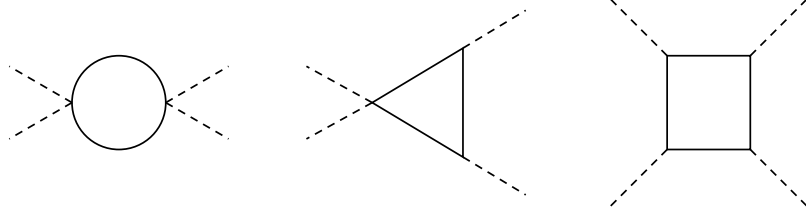


Figure 17.1: One-loop Feynman diagrams for the term in the quantum effective action that is quartic in a constant background gauge field A_α^μ . Here solid lines represent internal gauge, ghost, or matter lines; dashed lines indicate factors of A_α^μ . These three diagrams correspond to the three terms in Eq. (17.5.20).

$$[\text{tr} \ln \mathcal{M}]_{A^4} = \text{tr} \left\{ -\frac{1}{2} [\mathcal{M}_0^{-1} \mathcal{M}_2]^2 + [\mathcal{M}_0^{-1} \mathcal{M}_1]^2 \mathcal{M}_0^{-1} \mathcal{M}_2 - \frac{1}{4} [\mathcal{M}_0^{-1} \mathcal{M}_1]^4 \right\}. \quad (17.5.20)$$

(To see this, insert factors ϵ and ϵ^2 multiplying $\mathcal{M}_1$ and $\mathcal{M}_2$ in Eq. (17.5.19), differentiate $\text{tr} \ln \mathcal{M}$ four times with respect to ϵ , divide by $4!$ and set $\epsilon = 0$.) The $\mathcal{M}_0^{-1}$ factors here are just the usual propagators; for $\xi = 1$, these are

$$[\mathcal{M}_0^A(q)]_{\alpha\mu, \beta\nu}^{-1} = \delta_{\alpha\beta} \eta_{\mu\nu} (q^2 - i\epsilon)^{-1}, \quad (17.5.21)$$

$$[\mathcal{M}_0^\psi(q)]_{k\ell}^{-1} = [i\not{q} + m]_{k\ell}^{-1}, \quad (17.5.22)$$

$$[\mathcal{M}_0^\omega(q)]_{\alpha, \beta}^{-1} = \delta_{\alpha\beta} (q^2 - i\epsilon)^{-1}. \quad (17.5.23)$$

Indeed, the three terms in Eq. (17.5.20) just correspond to the three Feynman diagrams shown in Figure 17.1; the present method of calculation saves us from having to think about signs and combinatoric factors.

For the A loop, Eq. (17.5.15) gives for $\xi = 1$:

$$\begin{aligned} [\mathcal{M}_1^A(q)]_{\alpha\mu,\beta\nu} &= -2\eta_{\mu\nu}q^\lambda[\mathcal{A}_\lambda]_{\alpha\beta}, \\ [\mathcal{M}_2^A(q)]_{\alpha\mu,\beta\nu} &= \left[\eta_{\mu\nu}\mathcal{A}_\lambda\mathcal{A}^\lambda - \mathcal{A}_\nu\mathcal{A}_\mu - \mathcal{A}_\mu\mathcal{A}_\nu \right]_{\alpha\beta} + F_{\gamma\mu\nu}C_{\gamma\alpha\beta}, \end{aligned}$$

where $\mathcal{A}_\lambda$ is the matrix

$$[\mathcal{A}_\lambda]_{\alpha\beta} \equiv -iC_{\alpha\beta\gamma}A_{\gamma\lambda}$$

for which

$$\begin{aligned} [\mathcal{A}_\lambda, \mathcal{A}_\rho]_{\alpha\beta} &= -C_{\alpha\delta\gamma}C_{\delta\beta\epsilon}(A_{\gamma\lambda}A_{\epsilon\rho} - A_{\gamma\rho}A_{\epsilon\lambda}) \\ &= -(C_{\alpha\delta\gamma}C_{\delta\beta\epsilon} + C_{\alpha\delta\epsilon}C_{\delta\beta\gamma})A_{\gamma\lambda}A_{\epsilon\rho} \\ &= +C_{\alpha\delta\beta}C_{\delta\epsilon\gamma}A_{\gamma\lambda}A_{\epsilon\rho} = C_{\alpha\delta\beta}F_{\delta\rho\lambda}. \end{aligned}$$

The integrals have the structure

$$\begin{aligned} \int d^4q q^\mu q^\nu f(q^2) &= \frac{1}{4}\eta^{\mu\nu} \int d^4q q^2 f(q^2), \\ \int d^4q q^\mu q^\nu q^\rho q^\sigma f(q^2) &= \frac{1}{24}[\eta^{\mu\nu}\eta^{\rho\sigma} + \eta^{\mu\rho}\eta^{\nu\sigma} + \eta^{\mu\sigma}\eta^{\nu\rho}] \int d^4q (q^2)^2 f(q^2). \end{aligned}$$

We then find, for $\xi = 1$:

$$\begin{aligned} &\int d^4q \operatorname{tr} \left\{ [\mathcal{M}_0^A(q)^{-1}\mathcal{M}_2^A(q)]^2 \right\} \\ &\quad = 4\mathcal{I} \operatorname{tr}[\mathcal{A}^\lambda \mathcal{A}_\lambda \mathcal{A}^\eta \mathcal{A}_\eta] + 4\mathcal{I} C_{\gamma\alpha\beta}C_{\delta\alpha\beta}F_{\gamma\mu\nu}F_{\delta}^{\mu\nu}, \\ &\int d^4q \operatorname{tr} \left\{ [\mathcal{M}_0^A(q)^{-1}\mathcal{M}_1^A(q)]^2 \mathcal{M}_0^A(q)^{-1}\mathcal{M}_2^A(q) \right\} \\ &\quad = 4\mathcal{I} \operatorname{tr}[\mathcal{A}^\lambda \mathcal{A}_\lambda \mathcal{A}^\eta \mathcal{A}_\eta], \\ &\int d^4q \operatorname{tr} \left\{ [\mathcal{M}_0^A(q)^{-1}\mathcal{M}_1^A(q)]^4 \right\} \\ &\quad = \frac{8}{3}\mathcal{I} \operatorname{tr} \left[\begin{array}{l} 2\mathcal{A}^\lambda \mathcal{A}_\lambda \mathcal{A}^\eta \mathcal{A}_\eta \\ + \mathcal{A}^\lambda \mathcal{A}^\eta \mathcal{A}_\lambda \mathcal{A}_\eta \end{array} \right], \end{aligned}$$

where $\mathcal{I}$ is the divergent integral

$$\mathcal{I} \equiv \int d^4q [q^2 - i\epsilon]^{-2}, \quad (17.5.24)$$

whose significance is discussed below. Putting this together in Eq. (17.5.20), we have

$$\begin{aligned} \int d^4q [\operatorname{tr} \ln \mathcal{M}^A(q)]_{A^4} &= \frac{2}{3}\mathcal{I} \operatorname{tr}[\mathcal{A}^\lambda \mathcal{A}_\lambda \mathcal{A}^\eta \mathcal{A}_\eta - \mathcal{A}^\lambda \mathcal{A}^\eta \mathcal{A}_\lambda \mathcal{A}_\eta] \\ &\quad - 2\mathcal{I} C_{\gamma\alpha\beta}C_{\delta\alpha\beta}F_{\gamma\mu\nu}F_{\delta}^{\mu\nu}. \end{aligned}$$

Both terms are actually of the same form, and when combined yield

$$\int d^4q [\text{tr} \ln \mathcal{M}^A(q)]_{A^4} = -\frac{5}{3} \mathcal{I} C_{\gamma\alpha\beta} C_{\delta\alpha\beta} F_{\gamma\mu\nu} F_{\delta}^{\mu\nu}. \quad (17.5.25)$$

Now jumping ahead to the ghost loop, we see from Eq. (17.5.17) that

$$[\mathcal{M}_1^\omega(q)]_{\alpha\beta} = -2[\mathcal{A}^\lambda]_{\alpha\beta} q_\lambda, \quad (17.5.26)$$

$$[\mathcal{M}_2^\omega(q)]_{\alpha\beta} = [\mathcal{A}^\lambda \mathcal{A}_\lambda]_{\alpha\beta}. \quad (17.5.27)$$

We then find

$$\begin{aligned} \int d^4q \text{tr} \{ [\mathcal{M}_0^\omega(q)^{-1} \mathcal{M}_2^\omega(q)]^2 \} &= \mathcal{I} \text{tr} [\mathcal{A}^\lambda \mathcal{A}_\lambda \mathcal{A}^\eta \mathcal{A}_\eta], \\ \int d^4q \text{tr} \{ [\mathcal{M}_0^\omega(q)^{-1} \mathcal{M}_1^\omega(q)]^2 \mathcal{M}_0^\omega(q)^{-1} \mathcal{M}_2^\omega(q) \} &= \mathcal{I} \text{tr} [\mathcal{A}^\lambda \mathcal{A}_\lambda \mathcal{A}^\eta \mathcal{A}_\eta], \\ \int d^4q \text{tr} \{ [\mathcal{M}_0^\omega(q)^{-1} \mathcal{M}_1^\omega(q)]^4 \} &= \frac{2}{3} \mathcal{I} \text{tr} \left[\begin{array}{l} 2\mathcal{A}^\lambda \mathcal{A}_\lambda \mathcal{A}^\eta \mathcal{A}_\eta \\ + \mathcal{A}^\lambda \mathcal{A}^\eta \mathcal{A}_\lambda \mathcal{A}_\eta \end{array} \right]. \end{aligned}$$

Thus for the ghost loop the integral of the quantity (17.5.20) is

$$\begin{aligned} \int d^4q [\text{Tr} \ln \mathcal{M}^\omega(q)]_{A^4} &= \frac{1}{6} \mathcal{I} \text{Tr} [\mathcal{A}^\lambda \mathcal{A}_\lambda \mathcal{A}^\eta \mathcal{A}_\eta - \mathcal{A}^\lambda \mathcal{A}^\eta \mathcal{A}_\lambda \mathcal{A}_\eta] \\ &= \frac{1}{12} \mathcal{I} C_{\gamma\alpha\beta} C_{\delta\alpha\beta} F_{\gamma\mu\nu} F_{\delta}^{\mu\nu}. \end{aligned} \quad (17.5.28)$$

Finally, the vertices in the matter loop are

$$[\mathcal{M}_1^\psi(q)]_{k\ell} = -i(t_\alpha \mathcal{A}_\alpha)_{k\ell}, \quad [\mathcal{M}_2^\psi(q)]_{k\ell} = 0,$$

so here there is only one term in Eq. (17.5.20):

$$\int d^4q [\text{Tr} \ln \mathcal{M}^\psi(q)]_{A^4} = -\frac{1}{4} \int d^4q \text{Tr} \left\{ [(i\not{q} + m)^{-1} t_\alpha \mathcal{A}_\alpha]^4 \right\}.$$

We are interested in the ultraviolet-divergent part of this integral, so we may drop the mass (which is negligible for large q^ν) and write

$$\begin{aligned} \int d^4q [\text{Tr} \ln \mathcal{M}^\psi(q)]_{A^4} &= -\frac{1}{4} \text{Tr} \{ t_\alpha t_\beta t_\gamma t_\delta \} A_{\alpha\mu} A_{\beta\nu} A_{\gamma\rho} A_{\delta\sigma} \\ &\quad \times \int d^4q \frac{\text{Tr} \{ \not{q} \gamma^\mu \not{q} \gamma^\nu \not{q} \gamma^\rho \not{q} \gamma^\sigma \}}{(q^2 - i\epsilon)^8} \\ &= -\frac{\mathcal{I}}{96} \text{Tr} \{ t_\alpha t_\beta t_\gamma t_\delta \} A_{\alpha\mu} A_{\beta\nu} A_{\gamma\rho} A_{\delta\sigma} \\ &\quad \times \text{Tr} \left\{ \begin{array}{l} 2\gamma_\lambda \gamma^\mu \gamma^\lambda \gamma^\nu \gamma_\eta \gamma^\rho \gamma^\eta \gamma^\sigma \\ + \gamma_\lambda \gamma^\mu \gamma_\eta \gamma^\nu \gamma^\lambda \gamma^\rho \gamma^\eta \gamma^\sigma \end{array} \right\}. \end{aligned} \quad (17.5.29)$$

where $\mathcal{I}$ is the same divergent integral as in Eq. (17.5.24). To calculate the traces of Dirac matrices, we use the anticommutation relations of these matrices to write

$$\begin{aligned} \text{Tr} \left\{ \begin{aligned} &2\gamma_\lambda \gamma^\mu \gamma^\lambda \gamma^\nu \gamma_\eta \gamma^\rho \gamma^\eta \gamma^\sigma \\ &+ \gamma_\lambda \gamma^\mu \gamma_\eta \gamma^\nu \gamma^\lambda \gamma^\rho \gamma^\eta \gamma^\sigma \end{aligned} \right\} &= 8 \text{Tr}\{\gamma^\mu \gamma^\nu \gamma^\rho \gamma^\sigma\} \\ &\quad - 4 \text{Tr}\{\gamma^\nu \gamma^\mu \gamma^\rho \gamma^\sigma\} - 4 \text{Tr}\{\gamma^\mu \gamma^\rho \gamma^\nu \gamma^\sigma\} \\ &= -64\eta^{\mu\rho}\eta^{\nu\sigma} + 32\eta^{\mu\nu}\eta^{\rho\sigma} + 32\eta^{\mu\sigma}\eta^{\nu\rho}. \end{aligned}$$

Eq. (17.5.29) then gives

$$\begin{aligned} \int d^4q \left[\text{Tr} \ln \mathcal{M}^\psi(q) \right]_{A^4} &= \frac{1}{3} \mathcal{I} \text{Tr}\{[t_\alpha, t_\beta][t_\gamma, t_\delta]\} A_{\alpha\mu} A_{\beta\nu} A_\gamma^\mu A_\delta^\nu \\ &= -\frac{1}{3} \mathcal{I} F_{\gamma\mu\nu} F_\delta^{\mu\nu} \text{Tr}\{t_\gamma, t_\delta\}. \end{aligned} \quad (17.5.30)$$

Using Eqs. (17.5.25), (17.5.28), and (17.5.30) in Eq. (17.5.18) gives at last

$$\Gamma_{A^4}^{(1 \text{ loop})} = \frac{-i\mathcal{I}}{(2\pi)^4} \int d^4x F_{\gamma\mu\nu} F_\delta^{\mu\nu} \left[\left(\frac{5}{6} + \frac{1}{12} \right) C_{\gamma\alpha\beta} C_{\delta\alpha\beta} - \frac{1}{3} \text{Tr}\{t_\gamma, t_\delta\} \right], \quad (17.5.31)$$

where we have expressed the momentum-space delta-function in (17.5.14) as

$$\delta^4(p - p) = (2\pi)^{-4} \int d^4x \, 1. \quad (17.5.32)$$

It is important that the result turns out to depend on $A_{\alpha\mu}$ only through the field strength (17.5.11), as required by background gauge invariance.

Let us now (for the first time in this section) use the assumed simplicity of the gauge group and irreducibility of the matter field multiplet. In this case

$$C_{\gamma\alpha\beta} C_{\delta\alpha\beta} = g^2 C_1 \delta_{\gamma\delta}, \quad (17.5.33)$$

$$\text{Tr}\{t_\gamma, t_\delta\} = g^2 C_2 \delta_{\gamma\delta}, \quad (17.5.34)$$

where g is the common gauge coupling constant appearing as a factor in $C_{\gamma\alpha\beta}$ and t_γ , and C_1 and C_2 are numerical constants that characterize the gauge group and the representation of this group provided by the matter multiplet. For instance, in the original Yang–Mills theory the gauge group is $SU(2)$ (or equivalently $SO(3)$) and the structure constants are

$$C_{\gamma\alpha\beta} = g\epsilon_{\gamma\alpha\beta}$$

with α, β , and γ running over the values 1, 2, 3. Comparing with Eq. (17.5.33), we see that here

$$C_1 = 2.$$

Also in this theory the matter field forms a doublet with t_α given by $g/2$ times the usual Pauli matrix σ_α , so

$$C_2 = 1/2.$$

Somewhat more generally, for the group $SU(N)$ with n_f fermions in the defining representation, with a conventional normalization of generators, we have²

$$C_1 = N, \quad C_2 = n_f/2. \quad (17.5.35)$$

Returning now to the general case, Eqs. (17.5.33) and (17.5.34) give

$$\Gamma_{A^4}^{(1 \text{ loop})} = \frac{-ig^2 \mathcal{I}}{(2\pi)^4} \int d^4x F_{\gamma\mu\nu} F_\gamma^{\mu\nu} \left[\frac{11}{12} C_1 - \frac{1}{3} C_2 \right]. \quad (17.5.36)$$

That is, the infinite constant L_A in Eq. (??) is

$$L_A = \frac{4ig^2 \mathcal{I}}{(2\pi)^4} \left(\frac{11}{12} C_1 - \frac{1}{3} C_2 \right). \quad (17.5.37)$$

It remains to say a word about the interpretation of the divergent integral $\mathcal{I}$. First, before we try to integrate over the three-momentum $\mathbf{q}$, we can rotate the contour of integration of q^0 in Eq. (17.5.24) to the imaginary axis; as usual, the $-i\epsilon$ in the denominator forces us to rotate counterclockwise, so that $q^0 = iq^4$, with q^4 running from $-\infty$ to $+\infty$. The integral is then

$$\mathcal{I} = i \int_0^\infty \frac{2\pi^2 q^3 dq}{q^4}, \quad (17.5.38)$$

where q is now the magnitude of the Euclidean four-vector (q^1, q^2, q^3, q^4) . To go further, we evidently need some method of regulating the integral. The simplest way of dealing with the ultraviolet divergence is just to cut off the integral for q above a scale Λ . However, we also need a lower cut-off to deal with the infrared divergence. This is provided by the physics of the situation. If the momenta of the four vector particles is not zero, then a momentum flows through the internal lines of the diagrams, providing an infrared cut-off at the scale μ of these momenta. Similarly, if we evaluate the fourth variational derivative of $\Gamma[A]$ with respect to A not at $A = 0$ but at a finite A , then the propagators of the internal lines do not blow up at zero momenta, and we have an infrared cut-off at a scale $\mu \sim gA$. Either way, $\mathcal{I}$ takes the form

$$\mathcal{I} = 2\pi^2 i \int_\mu^\Lambda \frac{dq}{q} = 2\pi^2 i \ln \left(\frac{\Lambda}{\mu} \right) \quad (17.5.39)$$

²For $SU(3)$ the t_α are taken as $g/2$ times the Gell-Mann matrices λ_α used in Section 19.7, so that $C_{\alpha\beta\gamma} = (g/2)f_{\alpha\beta\gamma}$.

and hence

$$L_A = -\frac{g^2}{2\pi^2} \left(\frac{11}{12}C_1 - \frac{1}{3}C_2 \right) \ln \left(\frac{\Lambda}{\mu} \right) + O(g^4). \quad (17.5.40)$$

Eq. (??) then gives the renormalized coupling as

$$g_R = g \left[1 + \frac{g^2}{4\pi^2} \ln \left(\frac{\Lambda}{\mu} \right) \left(\frac{11}{12}C_1 - \frac{1}{3}C_2 \right) + O(g^4) \right]. \quad (17.5.41)$$

We note that while in quantum electrodynamics the radiative corrections discussed in Section 11.2 decrease the physical coupling g_R relative to the bare coupling g , in non-Abelian gauge theories they *increase* the physical coupling over the bare coupling, provided that the fermion multiplet is small enough so that $C_2 < 11C_1/4$. The importance of this point will be explored in Chapter 18.

Alternatively, we can deal with the ultraviolet divergence by the methods of dimensional regularization, discussed in Section 11.2. Here, in place of Eq. (17.5.39), we write

$$\mathcal{I} = i \int_0^\infty \frac{2\pi^2 q^{d-1} dq}{(q^2 + \mu^2)^2}, \quad (17.5.42)$$

where d is a complex dimensionality, allowed to approach 4 at the end of the calculation, and μ is an infrared cut-off, again taken of the order of the external momenta (or of the background fields times g). As long as d is complex with $\text{Re } d < 0$ and $\mu^2 > 0$, this has the finite value

$$\mathcal{I} = -i\pi^2 \left(\frac{d}{2} - 1 \right) \mu^{d-4} \frac{\pi}{\sin[(d/2 - 2)\pi]}.$$

Analytically continuing to $d \rightarrow 4$, this is

$$\mathcal{I} \longrightarrow -2i\pi^2 \left[\frac{1}{d-4} + \ln \mu + \dots \right], \quad (17.5.43)$$

where $\dots$ denotes finite μ -independent terms. Here we have

$$L_A = \frac{g^2}{2\pi^2} \left(\frac{11}{12}C_1 - \frac{1}{3}C_2 \right) \left(\frac{1}{d-4} + \ln \mu + \dots \right) + O(g^4)$$

and so

$$g_R = g \left[1 - \frac{g^2}{4\pi^2} \left(\frac{11}{12}C_1 - \frac{1}{3}C_2 \right) \left(\frac{1}{d-4} + \ln \mu + \dots \right) + O(g^4) \right]. \quad (17.5.44)$$

Note that the ultraviolet divergence here takes a different form, but the dependence on the infrared cut-off μ is the same. Eq. (17.5.44) will provide an important input to our discussion of asymptotic freedom in Section 18.7.